

MAGNETOMETER (03)

Aims

- A. To determine the horizontal component of the Earth's magnetic field and the magnetic moment of a given bar magnet using the deflection-and vibration magnetometers.
- B. To compare the magnetic moments of two bar magnets using the vibration magnetometer.

Materials and Apparatus

Two bar magnets, Deflection magnetometer, Vibration magnetometer, Meter ruler, Vernier caliper, Stopwatch and Electric balance.

Theory

Consider the case where a bar magnet is placed in the close vicinity of a magnetometer,

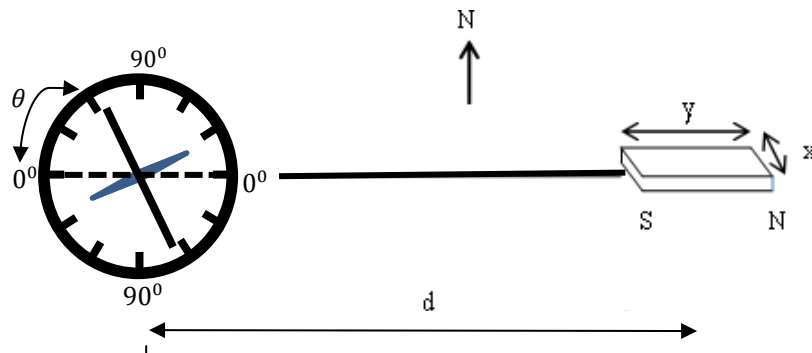


Fig 1.: The experimental setup

Where, d : distance between the midpoints of the deflection magnetometer and the bar magnet, y : geometrical length of the bar magnet, x : width of the bar magnet

- (i) The magnetic field at the center of the magnetometer due to the bar magnet,

$$B = \frac{\mu_0}{4\pi} \frac{2Md}{(d^2 - l^2)^2} = B_0 \tan \theta$$

$$\frac{M}{B_0} = \frac{(d^2 - l^2)^2 \tan \theta}{2d(\mu_0/4\pi)} \quad (1)$$

where,

M : magnetic moment of the bar magnet

$2l$: magnetic length of the bar magnet = $5y/6$

μ_0 : permeability of free space

θ : deflection of the magnetometer

B_0 : horizontal component of the Earth's magnetic field

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When the bar magnet is oscillated in the Earth's horizontal magnetic field, the period of oscillation is,

$$T = 2\pi \sqrt{\frac{I}{MB_0}}$$

$$MB_0 = \frac{4\pi^2 I}{T^2} \quad (2)$$

The moment of inertia of the bar magnet about the axis of oscillation;

$$I = \frac{m}{12} (x^2 + y^2)$$

where,

T : period of oscillation of the bar magnet

m : mass of the bar magnet

From equations (1) & (2),

$$M = \frac{2\pi}{T} (d^2 - l^2) \sqrt{\frac{I \tan \theta}{2(\mu_0/4\pi)d}} \quad (3)$$

$$B_0 = \frac{2\pi}{T(d^2 - l^2)} \sqrt{\frac{2I(\mu_0/4\pi)d}{\tan \theta}} \quad (4)$$

(ii) Period of oscillation when similar poles of the two magnets are,

Pointing in the same direction,

$$T_1 = 2\pi \sqrt{\frac{I_1 + I_2}{(M_1 + M_2)B_0}} \quad (5)$$

Pointing in the opposite direction,

$$T_2 = 2\pi \sqrt{\frac{I_1 + I_2}{|M_1 - M_2|B_0}} \quad (6)$$

From equations (5) & (6),

$$\frac{M_1}{M_2} = \frac{T_2^2 + T_1^2}{T_2^2 - T_1^2}$$

When two magnets are aligned perpendicular to each other, period of oscillation is,

$$T_3 = 2\pi \sqrt{\frac{I_1 + I_2}{B_0 \sqrt{M_1^2 + M_2^2}}} \quad (7)$$

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Procedure

Part (A) - Deflection Magnetometer

- a) Place the deflection magnetometer at the middle of the meter ruler and turn the dial of the deflection magnetometer so that the large needle points at 0° of the circular scale. At this position, the magnet in the magnetometer is pointing in the direction of Earth's magnetic field (B_0) and the meter ruler is along the direction perpendicular to B_0 .
- b) Place the bar magnet on the meter ruler at a distance d from the magnetometer (as in the Figure 1) and take the deflection of the needle.

Q: How do you determine the suitable range for d ?

Q: For a given deflection how many magnetometer readings can be obtained?

- c) Rotate the bar magnet by 180° by keeping d constant and obtain the deflection again.
- d) Repeat the above procedure for different values of d .

Q: How many different d values should be taken to get more accurate results?

- e) Repeat the above steps to the other side of the meter ruler.

Q: What is the advantage of repeating the measurements?

Part (B) - Vibration Magnetometer

- a) Place one of the bar magnets in the vibration magnetometer and rotate the magnetometer to align the magnet with the marked line at the bottom.

Q: Explain the purpose of this alignment.

- b) Give a small displacement to the bar magnet to perform torsional oscillations in a horizontal plane.
- c) Measure the time taken for suitable number of oscillations. Determine the period.

Q: How do you determine the number of oscillations to be considered to measure the period?

- d) Repeat the above steps in part B for the other magnet.
- e) Measure the mass, width, and length of the bar magnets.
- f) Show that the values satisfy the relationship,

$$\frac{2}{T_3^4} = \frac{1}{T_1^4} + \frac{1}{T_2^4}$$

Q: If the rectangular bar magnet is replaced by a magnet with a circular cross section, with the same length and the diameter equal to the width of the bar magnet, what effect does it have on the period?